

Eagle's Nest

Silverthorne · 617.2 SF timber frame · the standard

CHECK OR ACH	FINANCED	MONTHLY	SQUARE FEET	FOOTINGS	RAIL
\$122,615	\$131,844	\$1,530	617.2	8	62.2 LF

DESIGN READS

Frame	3 zones, 55'-2 3/8" along the house, 23'-0" at the deepest
Deck	62.2 LF of outer edge · 617.2 SF · 8'-0" above grade
Zone	C Deep wing: 12'-7 3/16" x 23'-0", wall set back 16'-0" · hot tub
Zone	B Dining terrace: 14'-7 3/16" x 9'-0", wall set back 2'-0"
Zone	A Lounge: 28'-0" x 7'-0"
Joists	57 4x10 Douglas Fir #1 timber (S4S, end grain sealed) @ 12" OC — 1,032 LF incl. rims, ledgers, blocking
Rims	single 4x10 rims
Beam	FLUSH 6x12 DF C: 12'-0 3/16", 2 posts at 1'-9 1/2" / 10'-9 11/16" (worst 6,513 lb)
Beam	DROP 6x12 DF C+B+A: 55'-1 3/8", 6 posts at 1'-6 1/2" / 12'-7 3/16" / 19'-10 13/16" / 27'-2 3/8" / 40'-5 1/8" / 53'-7 7/8" (worst 7,573 lb)
Posts	8 x 8x8 on 20" caissons 3'-6" deep with stone column bases
Load	110 psf (hot tub — 100 psf live) · engineered
Decking	TimberTech Vintage Coastline — 80 rows parallel to the house, picture frame & dividers in Vintage Dark Hickory, 1/8" gaps, Cortex
Rail	IRX 36" Matte Black: 9 bays, 10 posts, 62.2 LF + drink rail

- FLUSH 6x12 DF C: 12'-0 3/16" long, 2 posts at 1'-9 1/2" / 10'-9 11/16"
- DROP 6x12 DF C+B+A: 55'-1 3/8" long, 6 posts at 1'-6 1/2" / 12'-7 3/16" / 19'-10 13/16" / 27'-2 3/8" / 40'-5 1/8" / 53'-7 7/8"; pieces 12'-6 11/16" · 14'-7 3/16" · 13'-2 3/4" · 14'-8 3/4"
- B: 3'-0" cantilever past the drop beam exceeds L/4 of the 5'-8 1/2" back-span — engineered condition (owner-directed)
- A: 3'-0" cantilever past the drop beam exceeds L/4 of the 3'-8 1/2" back-span — engineered condition (owner-directed)
- A: field boards butt-joint: 2 pieces per row, stagger joints on alternating joists
- no fascia — the rim is the finished edge under a 1-1/2" board overhang (timber, end grain sealed)

- timbers unfinished — weather to gray; dark walnut oil is an option

SITE & CODE FLAGS internal

ENGINEER	SNOW	ground snow 80 psf > 40 psf — IRC R507 span tables do not apply; spans here are scaled estimates, a stamped design is required IRC R507.5/R507.6 (tables valid to 40 psf snow)
ENGINEER	ENGINEERING	hot tub — 100 psf live load; joists 12" OC, added beam/posts and footings under the tub, engineered design IRC R507 / ASCE 7
ENGINEER	ENGINEERING	joist: timber pre-engineering estimate (L/360 deflection governs) — stamped design required
ENGINEER	ENGINEERING	FLUSH 6x12 DF: timber pre-engineering estimate (bending governs) — stamped design required IRC Table R507.5
ENGINEER	ENGINEERING	DROP 6x12 DF: timber pre-engineering estimate (bending governs) — stamped design required IRC Table R507.5
ENGINEER	ENGINEERING	cantilever 3'-0" > L/4 of the 9'-0" back-span — owner-directed, in the stamped set IRC R507.5 / R507.6
ENGINEER	ENGINEERING	timber frame (DF #1 4x joists, 6x beams, 8x8 posts) is outside the IRC prescriptive tables — spans here are pre-engineering estimates; the stamped set governs sizes, post locations and caissons IRC R301.1.3
CODE	FIRE	WUI zone: Vintage is Ignition Resistant, Class A, WUI — keep the under-deck area clear of combustibles, 1-hr or noncombustible enclosure may be required IWUIC 504.7
CHECK	SETBACK	rear setback tight: 25' to the line vs 25' required — verify with a survey pin before layout zoning
CHECK	SETBACK	side setback tight: 0' to the line vs 0' required — verify with a survey pin before layout zoning
CHECK	LEDGER	verify the rim board behind the ledger is solid 2x/LVL/engineered rim (not I-joist web or a hollow band) before the ledger fasteners go in IRC R507.9.1.1
INFO	WIND	basic wind speed 115 mph — standard connectors
INFO	FIRE	IRX rail is noncombustible (Colorado Wildfire Resiliency Code practice: Class A decking, noncombustible rail, metal flashing at every wall)

THE ORDER Decks & Docks · GSEXT1 · ship to the job

Lumber

NET	ORDER	UNIT	ITEM	WHY
18	19	ea	4x10x10 Douglas Fir #1 timber (S4S, end grain sealed)	+1 cull (13 field joist · 2 PF joist · divider doubler · left rim ply · right rim ply)
59	60	ea	4x10x12 Douglas Fir #1 timber (S4S, end grain sealed)	+1 cull (13 joist upper (wall to the flush beam) · 13 joist lower (flush beam to the rim) · field joist, rail post block · blocking, field joist, rail post block · 17 blocking, field joist · 8 field joist · 2 PF joist · 2 divider doubler · left rim ply · right rim ply)
4	5	ea	4x10x16 Douglas Fir #1 timber (S4S, end grain sealed)	+1 cull (2 front rim ply · 2 ledger)
4	5	ea	4x10x20 Douglas Fir #1 timber (S4S, end grain sealed)	+1 cull (left rim ply · right rim ply · front rim ply · ledger)
5	5	ea	6x12x16 Douglas Fir #1 timber (S4S, end grain sealed)	exact (flush beam · 4 drop beam: FLUSH 6x12 DF C · DROP 6x12 DF C+B+A)
8	8	ea	8x8x8 Douglas Fir #1 timber (S4S, end grain sealed)	exact (yields 8 posts at ~5'-7 1/2" / 6'-4 3/4")

Footings

NET	ORDER	UNIT	ITEM	WHY
3	3	ea	Sonotube 20" x 12' (cut to length — 3 caissons per tube) — caissons 20" x 3'-6" + 6" above grade (frost 3'-4")	exact (8 caissons, 3 per tube)
120	120	bag	Quikrete 80# concrete mix	exact (15 bags per caisson)
7	7	ea	#4 rebar x 20' (4 verticals per caisson)	exact — engineer governs
7	7	ea	#3 rebar x 20' (ties)	exact — engineer governs
8	8	ea	5/8" x 8" HDG anchor bolt w/ nut + washer (cast-in)	exact
8	8	ea	Simpson 8x8 post base, BLACK powder-coat (Outdoor Accents APB88 or shop-coated ABU88Z)	exact
8	8	ea	Stone veneer column base 2'-0" x 2'-0" x 3'-0" — cement board wrap, lath, scratch coat, ledgerstone veneer (~24 SF) + 24" x 24" stone cap	exact (all 8 posts)

Hardware

NET	ORDER	UNIT	ITEM	WHY
89	89	ea	Simpson LUS410Z 4x10 hanger (ledgers + both faces of the flush beam), black shop-coat where visible	exact (joists at the ledgers + rim ends + both faces of the flush beam)
5	5	ea	Simpson HUC410 concealed hanger (doubled joists under dividers, hung rims), black shop-coat where visible	exact (doubled joists under the dividers, hung rims at the flush beam)
1	1	box	16d HDG ring-shank framing nail (2000 ct)	1636 needed — hanger headers, rim end-nailing, blocking toe-nails
2	2	box	Camo 9ga HDG hanger nail 1-1/2" 5 lb (~650)	1194 needed of ~1300

NET	ORDER	UNIT	ITEM	WHY
3	3	box	FastenMaster LedgerLOK 6" (50 ct) — 4x ledger to the house rim, 2 rows staggered 16"	107 needed
66	66	ea	Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden)	exact (every joist, doubler and rim at the drop beam)
4	4	ea	Simpson CCQ68SDS2.5 column cap 6x beam / 8x post, BLACK powder-coat (intermediate posts)	exact (intermediate posts)
4	4	ea	Simpson ECCQ68SDS2.5 end column cap, BLACK powder-coat (beam-end posts)	exact (beam-end posts)
1	1	lot	Black shop-coat for the visible hangers + touch-up paint	all visible hardware black
1	1	box	FastenMaster HeadLOK 6" (50 ct) — rail posts 4 ea	40 needed (10 rail posts x 4)
1	1	lot	Hot-tub bay: doubled joists + blocking under the tub footprint, per engineer	tub bay in zone C — RFI to the engineer

Flashing & waterproofing

NET	ORDER	UNIT	ITEM	WHY
2	2	roll	Grace Vycor Deck Protector 12" x 75' (wall membrane behind ledger + side rims)	71 LF of house wall with laps — nothing on a privacy wall
8	8	ea	Aluminum L-flashing 10' (over ledger + side rims)	house walls only
8	8	ea	26 ga galvanized wall flashing 12" x 10' (CWRC 404.3.2: noncombustible flashing at the deck/wall intersection, behind cladding)	WUI practice — house walls only
14	14	roll	Nichigo G-Tape 3035 4" x 65' black — tops of all 4x10 joists, rims, ledgers, beams	865' needed of 910'
2	2	roll	Nichigo G-Tape 3035 2" x 65' black — blocking, laps	82' needed of 130'

Decking

NET	ORDER	UNIT	ITEM	WHY
18	20	ea	TimberTech Vintage Coastline 1x6x12 Square Edge	+2 (C: 18 rows @ 11'-10 15/16" · C: 30 rows @ 12'-4 9/16" · B: 18 rows @ 13'-10 15/16" · A1: 14 rows @ 13'-7 7/8" · A2: 14 rows @ 12'-8 5/8")
76	78	ea	TimberTech Vintage Coastline 1x6x16 Square Edge	+2 (C: 18 rows @ 11'-10 15/16" · C: 30 rows @ 12'-4 9/16" · B: 18 rows @ 13'-10 15/16" · A1: 14 rows @ 13'-7 7/8" · A2: 14 rows @ 12'-8 5/8")
2	2	ea	TimberTech Vintage Coastline 1x6x16 Square Edge — rip boards at the house	exact (1.75" · 1.75" · 2.50" · 1.00" · 1.00" strips)
4	4	ea	TimberTech Vintage Dark Hickory 1x6x16 Square Edge — borders & dividers	exact (outer border 12'-7 3/16"-27'-2 3/8" 14'-1 7/16" · outer border 41'-2 3/8"-55'-3 7/8" 13'-7 3/4" · outer border 27'-2 3/8"-41'-2 3/8" 13'-6 1/4" · divider DA1 7'-1 1/2")
2	3	ea	TimberTech Vintage Dark Hickory 1x6x20 Square Edge — borders & dividers	+1 (outer border -1 1/2"-12'-7 3/16" 12'-8 11/16" + border end:right 7'-1 1/2" · divider DC/B 9'-1 1/2" + divider DB/A 7'-1 1/2")

Fasteners

NET	ORDER	UNIT	ITEM	WHY
9	9	box	Cortex for TimberTech PVC, Coastline plugs (350 ct / 100 SF)	2938 screws (2 per board at every bearing point, rips included) · 3150 supplied
1	1	box	Cortex for TimberTech PVC, Dark Hickory plugs — borders + dividers	152 screws

Rail

NET	ORDER	UNIT	ITEM	WHY
9	9	kit	TimberTech IRX Cable Rail kit 8' x 36" black (top rail, cable, fittings; cut to bay)	exact (9 bays, 6'-10 3/4" / 6'-2 3/8" / 6'-7" / 7'-0" / 7'-3 5/8" CTC)
10	10	kit	TimberTech IRX 36" post kit (post + cap + skirt) black	exact (1 CORNER, 2 END, 7 LINE)
4	5	ea	Drink rail — TimberTech Vintage Dark Hickory 1x6x16 Square Edge laid flat on the top rail, 62.2 LF, mitred at every turn	+1 (full-board line only, never scalloped)
9	9	kit	TimberTech drink-rail brackets, black — per 8' section (Build SOP: always TimberTech drink-rail brackets)	exact (one per bay)

Finish

NET	ORDER	UNIT	ITEM	WHY
1	1	gal	End-grain sealer (Anchorseal class) — every cut	428 cut ends — base: timbers unfinished, cuts sealed

Site

NET	ORDER	UNIT	ITEM	WHY
451.6	451.6	SF	Demo existing deck + haul-off	labor line — field verify the existing deck

FASTENER & CONNECTOR SCHEDULE

Ledger	4x10 ledger x 55'-2 3/8" total · LedgerLOK 6" 2 rows staggered 16" OC on 71.2 LF of house wall (ledgers + return walls; nothing on a privacy wall) · 107 fasteners
Blocking	116 x 4x10 blocks, (3) 16d toe-nails each end
Joist tape	G-Tape 4" on every 4x10 top (joists, rims, ledgers) and the beams; 2" on blocking
Field	Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (2938 field + 152 border/divider)
Rail	IRX 36" — 10 posts (1 CORNER, 2 END, 7 LINE), 9 bays at 6'-10 3/4" / 6'-2 3/8" / 6'-7" / 7'-0" / 7'-3 5/8" CTC (8' kits cut to bay), posts on the divider

lines, no bottom rail; HeadLOK 6" x 4 per post; drink rail board on TimberTech drink-rail brackets, mitred at every turn

▼ Cut list

MEMBER	STOCK	LENGTH	QTY	NOTE
Joists upper (C)	4x10	10'-5 1/2"	13	hung on the inner face of the flush beam
Joists lower (C)	4x10	11'-5 1/2"	13	hung on the outer face of the flush beam, bears the drop beam
Side rim plies (C)	4x10	22'-5"	2	single 4x rim
Front rim plies (C)	4x10	12'-7 3/16"	1	single 4x rim, end-nailed
Ledger (C)	4x10	12'-7 3/16"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking (C)	4x10	8 1/2"	60	5 rows at 3'-10 5/16", 7'-5 3/16", 14'-0", 17'-0", 20'-0" from the wall
Field boards run C	Vintage Coastline	11'-10 15/16"	18	boards run parallel to the house
Field boards run C	Vintage Coastline	12'-4 9/16"	30	boards run parallel to the house
Rip at the house (C)	Vintage Coastline	11'-10 15/16"	2	rip to 1.75"
Field joists (B)	4x10	8'-5"	13	@ 12" OC, crown up
Picture-frame joists (B)	4x10	8'-5"	2	centre 6 3/4" from each frame face
Divider doublers (B)	4x10	8'-5"	1	sister a joist on the divider line so both board edges bear
Side rim plies (B)	4x10	8'-5"	2	single 4x rim
Front rim plies (B)	4x10	14'-7 3/16"	1	single 4x rim, end-nailed
Ledger (B)	4x10	14'-7 3/16"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking (B)	4x10	8 1/2"	28	2 rows at 3'-1 3/4", 6'-0" from the wall
Field boards run B	Vintage Coastline	13'-10 15/16"	18	boards run parallel to the house
Rip at the house (B)	Vintage Coastline	13'-10 15/16"	1	rip to 2.50"
Field joists (A)	4x10	6'-5"	27	@ 12" OC, crown up
Picture-frame joists (A)	4x10	6'-5"	2	centre 6 3/4" from each frame face
Divider doublers (A)	4x10	6'-5"	2	sister a joist on the divider line so both board edges bear
Side rim plies (A)	4x10	6'-5"	2	single 4x rim
Front rim plies (A)	4x10	28'-0"	1	single 4x rim, end-nailed

MEMBER	STOCK	LENGTH	QTY	NOTE
Ledger (A)	4x10	28'-0"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking (A)	4x10	8 1/2"	28	1 row at 4'-0" from the wall
Field boards run A1	Vintage Coastline	13'-7 7/8"	14	boards run parallel to the house
Field boards run A2	Vintage Coastline	12'-8 5/8"	14	boards run parallel to the house
Rip at the house (A)	Vintage Coastline	13'-7 7/8"	2	rip to 1.00"
FLUSH 6x12 DF C beam	6x12	12'-0 3/16"	1	y -5'-0"; posts at 1'-9 1/2" / 10'-9 11/16"
Posts under FLUSH 6x12 DF C	8x8	6'-4 3/4"	2	field-measure each
DROP 6x12 DF C+B+A beam	6x12	55'-1 3/8"	4	y 4'-0"; posts at 1'-6 1/2" / 12'-7 3/16" / 19'-10 13/16" / 27'-2 3/8" / 40'-5 1/8" / 53'-7 7/8"; pieces 12'-6 11/16" · 14'-7 3/16" · 13'-2 3/4" · 14'-8 3/4", spliced over posts
Posts under DROP 6x12 DF C+B+A	8x8	5'-7 1/2"	6	field-measure each
Outer border -1 1/2"-12'-7 3/16"	Vintage Dark Hickory square	12'-8 11/16"	1	mitre at the corners; dividers on the doubled joist
Outer border 12'-7 3/16"-27'-2 3/8"	Vintage Dark Hickory square	14'-1 7/16"	1	mitre at the corners; dividers on the doubled joist
Outer border 27'-2 3/8"-41'-2 3/8"	Vintage Dark Hickory square	13'-6 1/4"	1	mitre at the corners; dividers on the doubled joist
Outer border 41'-2 3/8"-55'-3 7/8"	Vintage Dark Hickory square	13'-7 3/4"	1	mitre at the corners; dividers on the doubled joist
Border end:right	Vintage Dark Hickory square	7'-1 1/2"	1	mitre at the corners; dividers on the doubled joist
Divider dc/b	Vintage Dark Hickory square	9'-1 1/2"	1	mitre at the corners; dividers on the doubled joist
Divider db/a	Vintage Dark Hickory square	7'-1 1/2"	1	mitre at the corners; dividers on the doubled joist
Divider da1	Vintage Dark Hickory square	7'-1 1/2"	1	mitre at the corners; dividers on the doubled joist

THE STACK

internal

Materials (raw)	\$35,353.38
Materials × 1.15	\$40,656.39
Sales tax 4.00% on factored materials	\$1,626.26
Labor (rate card)	\$24,080.76

Work — carries the margin	\$66,363.40
General conditions at cost	\$7,200.00
Cost	\$73,563.40
Sell — check or ACH at 42.5% GM	\$122,615 · \$198.66/SF
Sell at the 40% floor	\$117,806
Financed	\$131,844 · \$1,530/mo at 6.99% × 10 yr
Gross profit	\$49,052
Engineering — outside the price, at cost	\$1,800–\$3,000

Labor

LINE	QTY	UNIT	RATE	EXT
Demolition — existing deck	451.6	SF	\$2.00	\$903.20
Caissons 20" x 3.5' (frost 40")	8	hole	\$150.00	\$1,200.00
Large footings 20" add-on	8	footing	\$150.00	\$1,200.00
Frame	617.2	SF	\$10.00	\$6,172.00
Decking (no fascia)	617.2	SF	\$8.00	\$4,937.60
Railing	62.2	LF	\$20.00	\$1,244.00
Cable rail add-on	62.2	LF	\$40.00	\$2,488.00
Drink rail add-on	62.2	LF	\$5.00	\$311.00
Extra beam set	1	set	\$200.00	\$200.00
Hot-tub bay framing + coordination with the tub set / electrician	1	ea	\$350.00	\$350.00
Stone column bases 2' x 2' x 3' — 8 posts (mason)	8	ea	\$615.62	\$4,924.96
Seal every cut end (base — no oil)	1	lot	\$150.00	\$150.00

General conditions — at cost, no margin

ITEM	AMOUNT	WHY
Project management & supervision — 20 hr	\$1,700	layout, holds, engineer + D&D coordination, QC gates
Mobilization / demobilization — trailer, tools, lift	\$800	Littleton to Silverthorne
811 utility locate + survey day	\$250	
Dumpster 20 yd — demo + construction	\$650	
Portable toilet	\$200	
Material delivery — D&D Denver to Silverthorne, 2 drops	\$700	boom truck for the timbers
Equipment — 20" auger rig, lift day for the 6x12s	\$600	
Site protection — Ram Board, plywood paths, tarps, tree protection	\$250	

ITEM	AMOUNT	WHY
Mountain travel / per diem — 15 crew-days	\$1,500	
Final clean, magnet sweep x 2	\$150	
Contingency	\$400	

What the price includes — sell values

ITEM	VALUE
Timber-frame deck: caissons, Douglas fir frame (end grain sealed), Coastline decking, IRX cable rail, black hardware, site protection, delivery, cleanup	\$98,035
Stone column bases at all 8 posts	\$14,888
Dark Hickory drink rail on the cable rail	\$2,492
Site & project services	\$7,200
Total, check or ACH	\$122,615
Total, financed	\$131,844

Options — full installed deltas

OPTION	COST Δ	CHECK / ACH	FINANCED
Fortress Evolution steel frame	+1,132	+1,969	+2,117
Dark walnut oil finish on all timbers	+2,560	+4,452	+4,787
Concrete pad under the deck	+10,210	+17,757	+19,094
Privacy wall, Hardie board & batten — 23'	+4,369	+7,597	+8,169
Privacy wall, aluminum slats — 23'	+7,327	+12,743	+13,702
TimberTech Fulton Rail in place of the IRX cable rail	-5,364	-9,329	-10,031
No stone column bases	-8,561	-14,888	-16,009
No drink rail	-1,433	-2,492	-2,680

THE QUOTE homeowner-facing · facts only

QUOTE · PRE-ENGINEERING

EAGLE'S NEST 617 SF DECK

DF #1 timber frame — 8x8 posts, 6x12 beams, 4x10 joists at 12" — on 8 caissons with stone column bases. TimberTech Vintage Coastline decking with a Dark Hickory border, IRX cable rail with a Dark Hickory drink rail, hot-tub bay in zone C.

617.2 SF · 3 zones · **8'-0"** above grade, engineered · **80 psf** snow design

Prepared for Dave & Julie Wirth · 9 Eagle's Nest Court, Silverthorne

PRICE

Total investment \$131,844 — financed, no money down

- 12 months no payments, no down payment
- 18 months no payments, no down payment
- 6.99% for 10 years — about \$1,530/month

Check or ACH: \$122,615 (7% savings)

WHAT THE PRICE INCLUDES

ITEM	VALUE
Timber-frame deck: caissons, Douglas fir frame (end grain sealed), Coastline decking, IRX cable rail, black hardware, site protection, delivery, cleanup	\$98,035
Stone column bases at all 8 posts	\$14,888
Dark Hickory drink rail on the cable rail	\$2,492
Site & project services	\$7,200
Total, check or ACH	\$122,615
Total, financed	\$131,844

Engineering — \$1,800 to \$3,000. Stamped drawings by a Colorado engineer (80 psf snow, timber frame, caissons, hot-tub bay). GSX provides the engineer and bills the engineering at cost, with no markup. You own the stamped set. It is separate from the deck price above.

INCLUDED — WHAT IT IS

ELEMENT	SELECTION
Decking	TimberTech PVC Vintage, Coastline; picture frame and dividers in Vintage Dark Hickory. Square-shoulder boards, 1/8" gaps, no fascia. Ignition Resistant, Class A, WUI; Limited Lifetime product / 50-yr fade & stain.
Railing	TimberTech IRX cable rail, Matte Black, 36", no bottom rail; Dark Hickory drink rail on top. 62.2 LF.
Timbers	Douglas fir #1 8x8 posts, 6x12 beams, 4x10 joists at 12"; unfinished, end grain sealed. Tub bay doubled.
Finish	Base: unfinished Douglas fir, which weathers gray. Dark walnut oil finish is priced in Options.

ELEMENT SELECTION

Column bases	Stone veneer 2'x2' x 3' with a 24" stone cap at all 8 posts.
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Foundation	20" caissons 3'-6" deep with stone column bases (frost depth 3'-4").
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Design load	80 psf snow, stamped by a Colorado engineer. Colorado Wildfire Resiliency Code practice: Class A decking, noncombustible rail, metal flashing at every wall.
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Hardware	All visible hardware black powder-coat. Cortex fasteners with color-matched plugs.
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PLAN

3 zones · 617.2 SF · 55'-2 3/8" along the house · 8'-0" above grade · 62.2 LF Matte Black rail

- **C · Deep wing** — 12'-7 3/16" x 23'-0", hot-tub bay against the house, no rail on the wall side.
- **B · Dining terrace** — 14'-7 3/16" x 9'-0".
- **A · Lounge** — 28'-0" x 7'-0".

Zones are named left to right standing in the yard facing the house.

SCHEDULE

1. Engineering & ordering (2-3 weeks). Stamped drawings, materials ordered, samples approved.
2. Day 1 — protection, utilities located (811), old deck out.
3. Days 2-3 — layout, 8 caissons augered and poured.
4. Days 4-5 — concrete cure.
5. Days 6-9 — posts and beams set (first coat of oil if the finish option is taken).
6. Days 10-14 — timber frame, flashing at every wall, tub bay doubled.
7. Days 15-17 — decking, borders first.
8. Days 18-19 — IRX rail and drink rail. Owner's electrician and plumber finish the tub circuit.
9. Day 20 — stone bases and caps; end of construction.

OPTIONS — UPGRADES AND DOWNGRADES

- Fortress Evolution steel frame — add \$2,117 (\$1,969 check/ACH) — Fortress Evolution galvanized-steel joists in place of the 4x10 Douglas fir joists; material difference only — Henry to quote
- Dark walnut oil finish on all timbers — add \$4,787 (\$4,452 check/ACH)
- Concrete pad under the deck — add \$19,094 (\$17,757 check/ACH) — 4" broom-finish slab on the 617 SF footprint, poured after the caissons — owner-set \$17,757
- Privacy wall, Hardie board & batten — 23' — add \$8,169 (\$7,597 check/ACH)
- Privacy wall, aluminum slats — 23' — add \$13,702 (\$12,743 check/ACH)
- TimberTech Fulton Rail in place of the IRX cable rail — deduct \$10,031 (\$9,329 check/ACH)

- No stone column bases — deduct \$16,009 (\$14,888 check/ACH)
- No drink rail — deduct \$2,680 (\$2,492 check/ACH)

NEXT STEP

Engineering first. This is a quote: the numbers depend on the stamped engineering, so the first step is engineering. GSX provides the Colorado engineer and supplies the design package; you pay GSX the engineer's cost at no markup and you own the stamped set. When the stamped set is in hand, GSX issues a fixed-price proposal on these numbers with the payment terms.

Quote, not a contract: a fixed-price proposal follows the stamped engineering. Quote numbers hold 30 days. Renders and plan are design intent; the stamped engineering set governs structure, dimensions and locations.

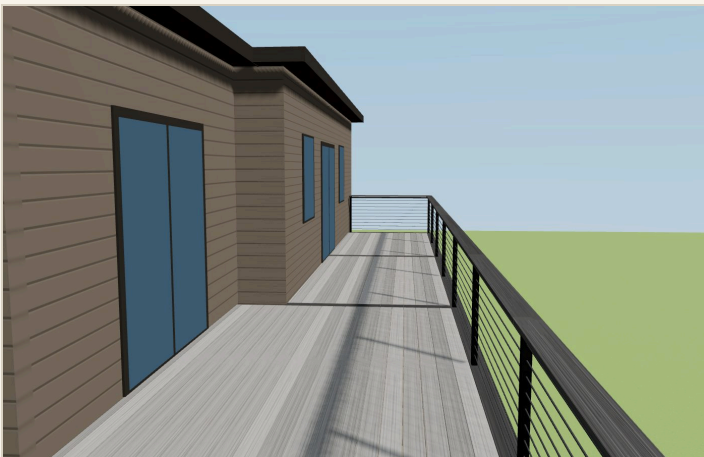
RENDERS 3D model · design intent · timbers shown with the optional oil finish



From the yard



From the front corner



Standing on the deck

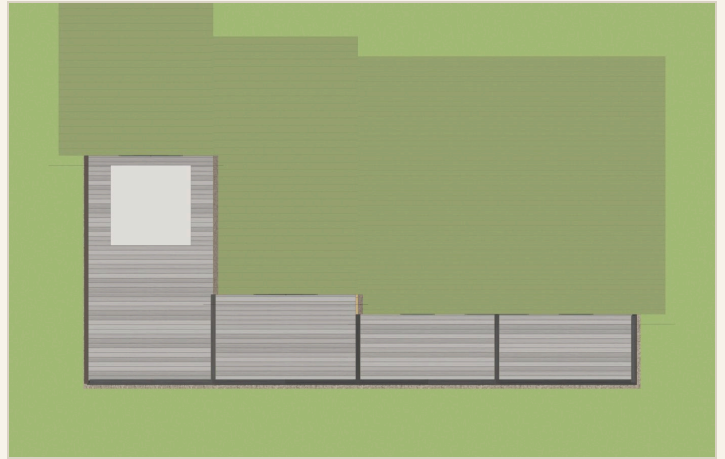


Under the deck — posts, beams, footings

ISOMETRICS



Isometric from the stair corner



Plan from above

EXPLODED



Exploded — every layer

G-001 General notes · design criteria

G-001

GENERAL NOTES · DESIGN CRITERIA

Loads, codes, structure summary, sheet index, site — the permit-set cover

Eagle's Nest 617 SF deck · 9 Eagle's Nest Court, Silverthorne

GSX · Jade Pender 303-550-9558 · September 16, 2026 · Rev 0 FOR REVIEW

PROJECT

Eagle's Nest 617 SF deck
9 Eagle's Nest Court, Silverthorne CO 80498
Prepared for Dave & Julie Wirth
617.2 SF · 8'-0" above grade · 3 zones

DESIGN CRITERIA

Deck live load 40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf
Ground snow 80 psf (ground; deck design uses the larger of snow and live) — snow governs
Dead load 10 psf deck
Roof snow (cover) —
Design total 110 psf on the deck (hot-tub bay 110 psf)
Wind 115 mph Vult, Exposure C, Risk Cat. II (ASCE 7-16) — confirm special wind region
Frost depth 3'-4" (footings bear below)
Soil bearing 2,000 psf presumptive (IRC R401.4.1) — no soils report
Seismic B
Wildfire (WUI) Ignition-resistant construction: Class A / WUI-listed decking (TimberTech Advanced PVC), noncombustible rail (steel), fiber-cement skirt, soffit and fascia, 26 ga flashing at every wall, Class A roof on the cover; under-deck kept clear of combustibles (Colorado Springs WUI 2021 IRC as adopted by Silverthorne; IRC R507 decks; ASCE 7-16 loads; IWUIC where mapped; timber members outside the tables: stamped design
Codes 2021 IRC as adopted by Silverthorne; IRC R507 decks; ASCE 7-16 loads; IWUIC where mapped; timber members outside the tables: stamped design
Engineering Stamped structural set by a Colorado PE required — the stamped set governs

STRUCTURE

Footings 20" caissons 3'-6" deep with stone column bases
Posts 8 x 8x8
Beams FLUSH 6x12 DF C: 12'-0 3/16", 2 posts at 1'-9 1/2" / 10'-9 11/16" (worst 6,513 lb); DROP 6x12 DF C+B+A: 55'-1 3/8", 6 posts at 1'-6 1/2" / 12'-7 3/16" / 19'-10 13/16" / 27'-2 3/8" / 40'-5 1/8"
Joists 57 4x10 Douglas Fir #1 timber (S4S, end grain sealed) @ 12" OC — 1,032 LF incl. rims, ledgers, blocking
Rims single 4x10 rims
Ledger 4x10 ledger x 55'-2 3/8" total · LedgerLOK 6" 2 rows staggered 16" OC on 71.2 LF of house wall (ledgers + return walls; nothing on a privacy wall) · 107 fasteners
Decking TimberTech Vintage Coastline — 80 rows parallel to the house, picture frame & dividers in Vintage Dark Hickory, 1/8" gaps, Cortex
Rail IRIX 36" Matte Black: 9 bays, 10 posts, 62.2 LF + drink rail

SHEET INDEX

G-001 General notes · design criteria
A-001 Isometric · renders
S-100 Foundation plan
S-101 Framing plan
A-101 Decking plan
A-201 Railing plan + elevation
A-301 Section + front elevation
D-501 Details
T-400 Build steps (build set)
T-700 Order + schedules (takeoff)

SITE

Rear / front lot line: deck edge 25' from the line, 25' required
Side lot line: deck edge 0' from the line, 0' required
Summit County criteria by choice (80 psf roof snow, 40" frost, 115 mph, seismic B); 2025 CWRC Class 2 practice; NO permit (owner election 6 Sep 2026). Front lot line 25' off the deck edge; the west lot line is the privacy wall (deck sits at it).

FLAGS (INTERNAL — remove before the permit submittal)

ENGINEER ground snow 80 psf > 40 psf — IRC R507 span tables do not apply;...
ENGINEER hot tub — 100 psf live load; joists 12" OC, added beam/posts and...
ENGINEER joist: timber pre-engineering estimate (L/360 deflection governs)...
ENGINEER FLUSH 6x12 DF: timber pre-engineering estimate (bending governs)...
ENGINEER DROP 6x12 DF: timber pre-engineering estimate (bending governs) —...
ENGINEER cantilever 3'-0" > L/4 of the 9'-0" back-span — owner-directed...
ENGINEER timber frame (DF #1 4x joists, 6x beams, 8x8 posts) is outside...
CODE WUI zone: Vintage is Ignition Resistant, Class A, WUI — keep the...

No scale

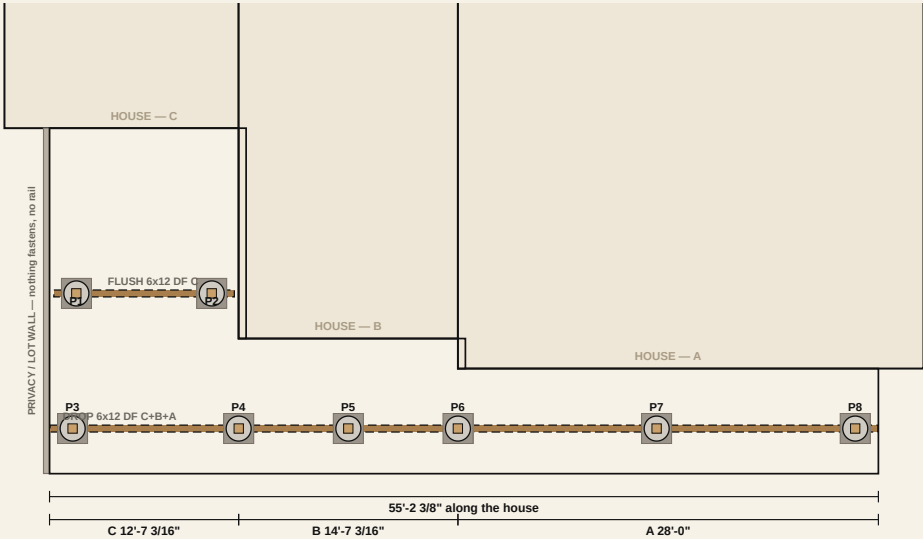
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale. Oil finish is an option; base timbers unfinished, end grain sealed.

FOOTING SCHEDULE

20" drilled caisson x 3'-6" + 6" above
grade, (4) #4 verticals + #3 ties, 5/8" x 8"
cast-in anchor

ID	X	Y	ZONE	LOAD
P1	1'-9 1/2"	-5'-0"	C	6,513 lb
P2	10'-9 11/16"	-5'-0"	C	6,513 lb
P3	1'-6 1/2"	4'-0"	C	5,799 lb
P4	12'-7 3/16"	4'-0"	C	7,573 lb
P5	19'-10 13/16"	4'-0"	B	3,846 lb
P6	27'-2 3/8"	4'-0"	B	5,408 lb
P7	40'-5 1/8"	4'-0"	A	5,779 lb
P8	53'-7 7/8"	4'-0"	A	3,545 lb

x from the left frame face facing the house; y from the reference wall.
Frost 3'-4" · soil 2,000 psf · worst post 7,573 lb



HOUSE — C

HOUSE — B

HOUSE — A

PRIVACY / LOT WALL — nothing fastens, no rail

C: 11 field + 2 PF 4x10 @ 12" OC · 22'-5"

B: 13 field + 2 PF 4x10 @ 12" OC · 8'-4"

A: 27 field + 2 PF 4x10 @ 12" OC · 6'-5"

doubled joist · DC/B

doubled joist · DB/A

doubled joist · DA1

55'-2 3/8"

FRAMING NOTES

57 4x10 Douglas Fir #1 timber (S4S, end grain sealed) @ 12" OC — 1,032 LF incl. rims, ledgers, blocking

single 4x10 rims

4x10 ledger x 55'-2 3/8" total · LedgerLOK 6" 2 rows staggered 16" OC on 71.2 LF of house wall (ledgers + return

116 x 4x10 blocks, (3) 16d toe-nails each end

G-Tape 4" on every 4x10 top (joists, rims, ledgers) and the beams; 2" on blocking

FLUSH 6x12 DF C: 12'-0 3/16", 2 posts at 1'-9 1/2" / 10'-9 11/16" (worst 6,513 lb); DROP 6x12 DF C+B+A: 55'-1 3/8", 6

Connectors:

Simpson LUS4102 4x10 hanger (ledgers + both faces of the flush beam), black shop-coat where visible: 89

Simpson HUC410 concealed hanger (doubled joists under dividers, hung rims), black shop-coat where visible: 5

Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden): 66

Simpson CCQ68SDS2.5 column cap 6x beam / 8x post, BLACK powder-coat (intermediate posts): 4

Simpson ECCQ68SDS2.5 end column cap, BLACK powder-coat (beam-end posts): 4

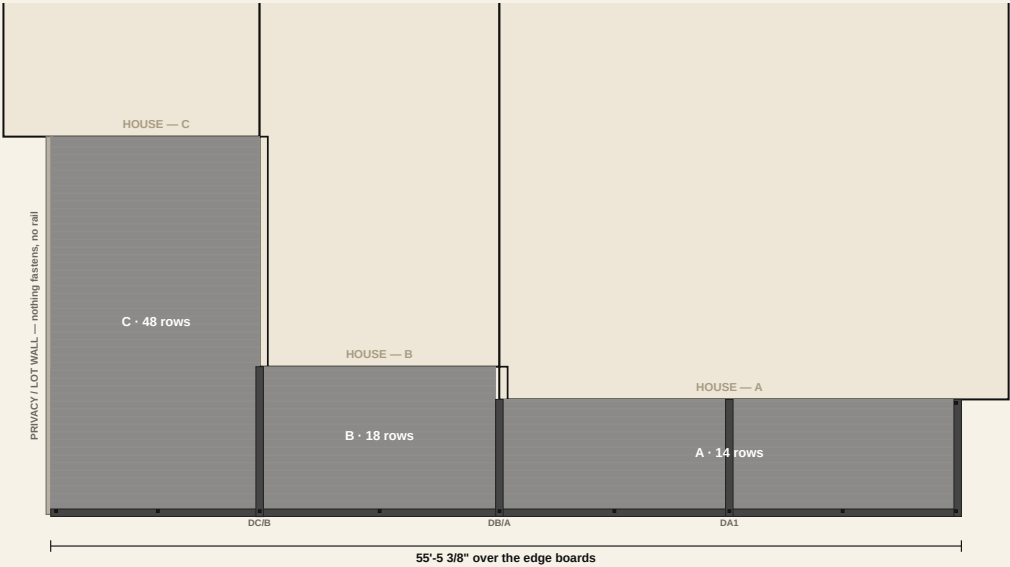
A-101 Decking plan

A-101 DECKING PLAN

Every board, border and divider as it lays; rail posts shown for the pockets

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DECKING

Field: TimberTech Vintage Coastline, 5.5" x 1", 1/8" gaps

Border & dividers: Vintage Dark Hickory

Boards run parallel to the house; edge overhang 1 1/2" — no fascia, rim exposed

Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (2938 field + 152 border/divider)

Rows per zone: C 48 · B 18 · A 14 (rip at the house)

A-201 Railing plan + elevation

A-201

RAILING PLAN + ELEVATION

Post IDs and types (E end · L line · C corner), bay lengths, outer-edge elevation

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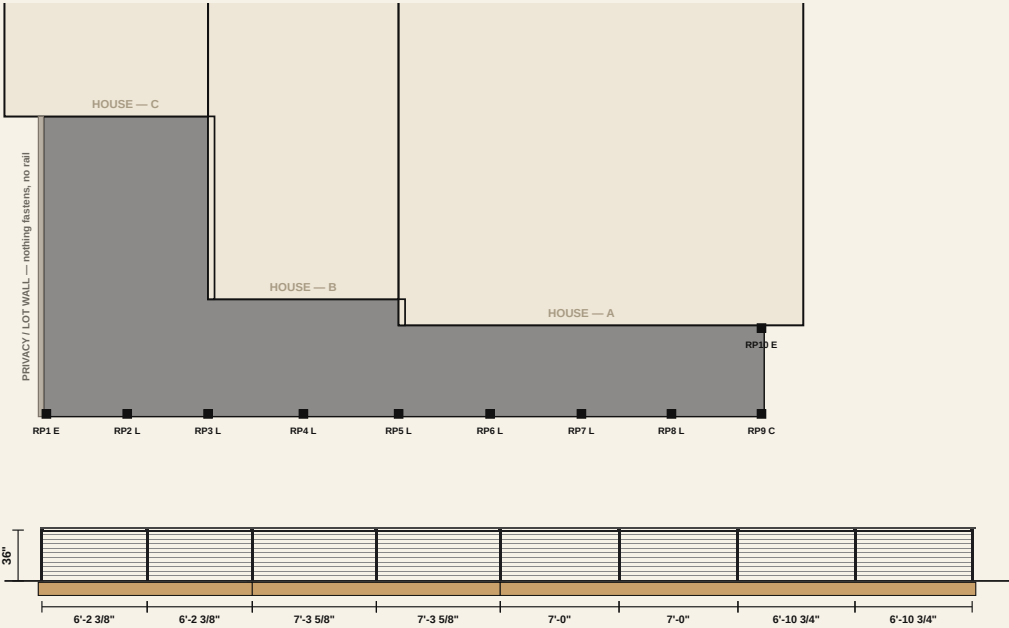
RAILING

IRX 36" Matte Black: 9 bays, 10 posts,
62.2 LF + drink rail

IRX 36" — 10 posts (1 CORNER, 2 END, 7
LINE), 9 bays at 6'-10 3/4" / 6'-2 3/8"
/ 6'-7" / 7'-0" / 7'-3 5/8" CTC (8' kits
cut to bay), posts on the divider lines,
no bottom rail: HeadLOK 6" x 4 per post;
drink rail board on TimberTech drink-

Bays: front:C+B+A 6'-2 3/8" ·
front:C+B+A 6'-2 3/8" · front:C+B+A 7'-3
5/8" · front:C+B+A 7'-3 5/8" ·
front:C+B+A 7'-0" · front:C+B+A 7'-0" ·
front:C+B+A 6'-10 3/4" · front:C+B+A
6'-10 3/4" · end:right 6'-7"

Guard 36" min above the deck; openings
under 4" (IRC R312). Rail LF for labor =
deck-edge LF of actual rail.



Plan above · elevation of the outer edge below

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale. Oil finish is an option; base timbers unfinished, end grain sealed.

A-301 Section + front elevation

A-301

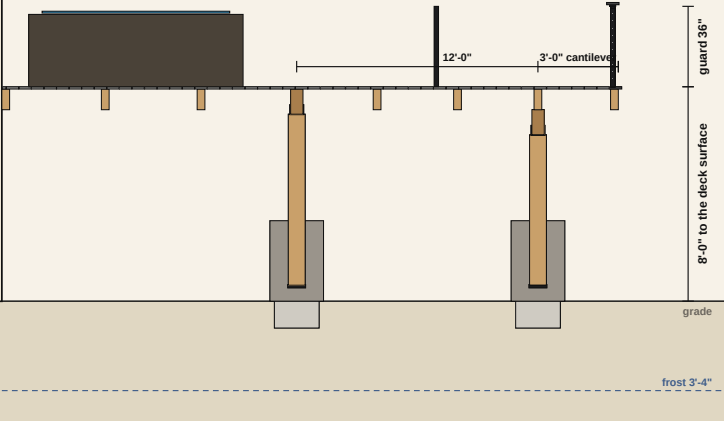
SECTION + FRONT ELEVATION

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

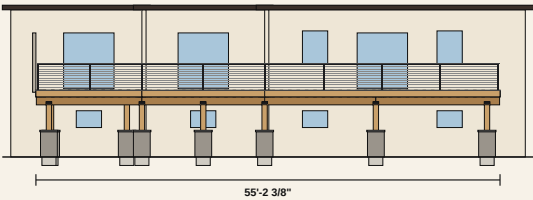
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SECTION through zone C at 6'-3 5/8" — looking toward the left end



FRONT ELEVATION — from the yard



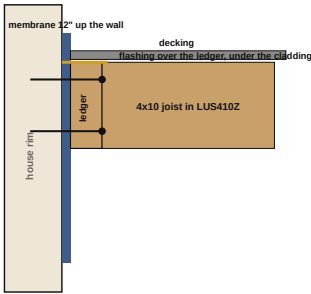
SECTION NOTES

Deck surface 8'-0" above grade; guard 36" where the drop exceeds 30"
FLUSH 6x12 DF C: 12'-0 3/16", 2 posts at 1'-9 1/2" / 10'-9 11/16" (worst 6,513 lb); DROP 6x12 DF C+B+A: 55'-1 3/8", 6 posts at 1'-6 1/2" / 12'-7 3/16" / 19'-10 13/16" / 27'-2 3/8"
Posts 8 x 8x8 on 20" caissons 3'-6" deep with stone column bases
4x10 ledger x 55'-2 3/8" total · LedgerLOK 6" 2 rows staggered 16" OC on 71.2 LF of house wall (ledgers + return walls; nothing on a privacy wall) · 107 fasteners
G-Tape 4" on every 4x10 top (poists, rims, ledgers) and the beams; 2" on blocking

Section left · elevation right · not to a common scale

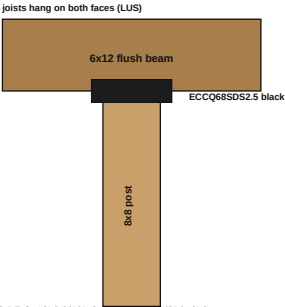
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale. Oil finish is an option; base timbers unfinished, end grain sealed.

1 LEDGER AT THE HOUSE



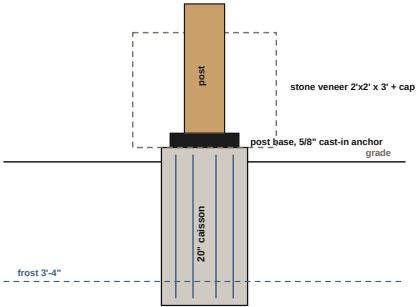
4x10 ledger x 55'-2 3/8" total · LedgerLOK 6" 2 rows staggered 16" OC on 71.2 LF of house wall (ledgers + return walls; nothing on a privacy wall) · 107 fasteners · 26 ga wall flashing (WUJ)

2 POST · BEAM · CAP

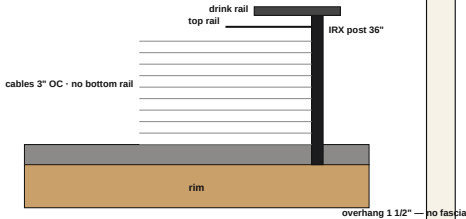


FLUSH 6x12 DF C: 12'-0 3/16", 2 posts at 1'-9 1/2" / 10'-9 11/16" (worst 6.513 lb); DROP 6x12 DF C+B+A: 55'-1 3/8", 6 posts at 1'-6 1/2" / 12'-7 3/16"

3 FOOTING · POST BASE

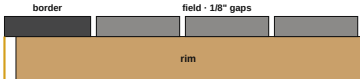


4 RAIL POST · GUARD



IRX 36" — 10 posts (1 CORNER, 2 END, 7 LINE), 9 bays at 6'-10 3/4" / 6'-2 3/8" / 6'-7" / 7'-0" / 7'-3 5/8" CTC (8" kits cut to bay), posts on the divider lines, no bottom rail, HeadLOK 6" x 4 per post; drink rail board on TimberTech drink-rail brackets, mitred at every turn

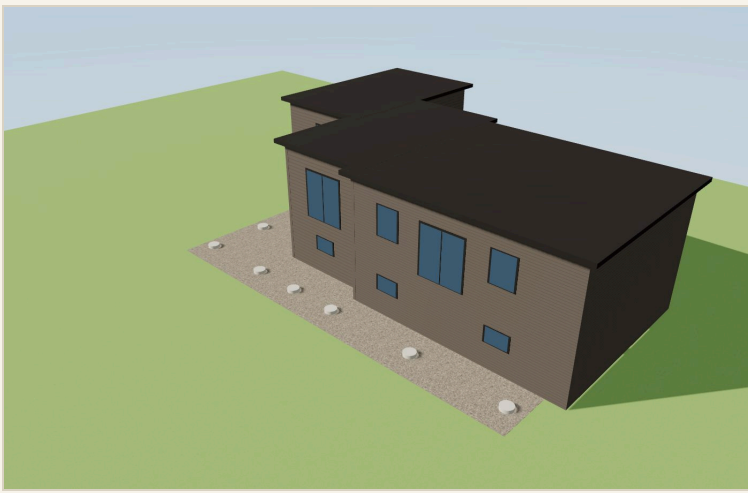
5 DECK EDGE · BORDER



Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (2938 field + 152 border/divider)

6 HOT-TUB BAY

Tub bay in zone C: 8'-0" square against the house — sister a joist to every joist across the bay, block between, design 100 psf; engineer confirms. Owner's electrician: 50A GFCI circuit + disconnect.



Step 1 • Footings — 8 caissons 20" x 3.5' below frost

T-400 • Step 1 • Footings

GOES IN

- (3) Sonotube 20" x 12' (cut to length — 3 caissons per tube) — caissons 20" x 3'-6" + 6" above grade (frost 3'-4")
- (120) Quikrete 80# concrete mix
- (7) #4 rebar x 20' (4 verticals per caisson)
- (7) #3 rebar x 20' (ties)
- (8) 5/8" x 8" HDG anchor bolt w/ nut + washer (cast-in)
- (8) Simpson 8x8 post base, BLACK powder-coat (Outdoor Accents APB88 or shop-coated ABU88Z)

HOW

- Call 811 and locate before any hole or pin. Strip the existing deck first.
- Pull the beam lines from the house wall: FLUSH 6x12 DF C at -5'-0" from the reference wall, posts at 1'-9 1/2" / 10'-9 11/16"; DROP 6x12 DF C+B+A at 4'-0" from the reference wall, posts at 1'-6 1/2" / 12'-7 3/16" / 19'-10 13/16" / 27'-2 3/8" / 40'-5 1/8" / 53'-7 7/8".
- Auger 8 holes 20" x 3'-6" (frost 3'-4"), tube to 6" above grade, cage in, pour, set the 5/8" anchors wet. Cure 2 days.

CHECK

- Footings on the layout within 1/2"; bearing on undisturbed soil
 - Frost depth met
 - Photos before backfill
-

T-401 · Step 2 · Posts

GOES IN

- (8) 8x8x8 Douglas Fir #1 timber (S4S, end grain sealed)
- (8) 5/8" x 8" HDG anchor bolt w/ nut + washer (cast-in)
- (8) Simpson 8x8 post base, BLACK powder-coat (Outdoor Accents APB88 or shop-coated ABU88Z)
- (1) End-grain sealer (Anchorseal class) — every cut

HOW

- 8x8 posts on the bases; cut so the beam top lands at the joist bottom (drop beam) or the joist top (flush beam): FLUSH 6x12 DF C posts 6'-4 3/4" / DROP 6x12 DF C+B+A posts 5'-7 1/2" — field-measure each.
- Plumb both ways, brace, then the caps go on with the beams.
- Seal every cut end before it goes up.

CHECK

- Posts plumb to a string; tops on the beam line
- Every cut end sealed



Step 2 · Posts — 8x8 Douglas fir #1 on black bases

T-402 · Step 3 · Stone column bases

GOES IN

- (8) Stone veneer column base 2'-0" x 2'-0" x 3'-0" — cement board wrap, lath, scratch coat, ledgerstone veneer (~24 SF) + 24" x 24" stone cap

HOW

- Mason after the posts and before the beams or after the frame — owner's call on access. Cement board wrap, lath, scratch, ledgerstone, 24" cap.
- Keep the cap 1/2" clear of the post; caulk the joint.

CHECK

- Stone plumb and square to the deck; caps level
- Sample color approved by the homeowner



Step 3 · Stone column bases — 2'x2' x 3' ledgerstone with 24" caps

T-403 · Step 4 · Beams

GOES IN

- (5) 6x12x16 Douglas Fir #1 timber (S4S, end grain sealed)
- (4) Simpson CCQ68SDS2.5 column cap 6x beam / 8x post, BLACK powder-coat (intermediate posts)
- (4) Simpson ECCQ68SDS2.5 end column cap, BLACK powder-coat (beam-end posts)

HOW

- FLUSH 6x12 DF C: 12'-0 3/16" one piece, 2 posts @ 9'-0 3/16" max
- DROP 6x12 DF C+B+A: 55'-1 3/8" in 4 pieces spliced over posts, 6 posts @ 13'-2 3/4" max
- Crown up, strung straight, caps screwed off with the SDS screws in the box.

CHECK

- Beam top = joist bottom (drop) / joist top (flush)
- Splices land on a post; caps fully fastened



Step 4 · Beams — FLUSH 6x12 DF C (2 posts) · DROP 6x12 DF C+B+A (6 posts)

T-404 · Step 5 · Frame

GOES IN

- (18) 4x10x10 Douglas Fir #1 timber (S4S, end grain sealed)
- (59) 4x10x12 Douglas Fir #1 timber (S4S, end grain sealed)
- (4) 4x10x16 Douglas Fir #1 timber (S4S, end grain sealed)
- (4) 4x10x20 Douglas Fir #1 timber (S4S, end grain sealed)
- (89) Simpson LUS410Z 4x10 hanger (ledgers + both faces of the flush beam), black shop-coat where visible
- (5) Simpson HUC410 concealed hanger (doubled joists under dividers, hung rims), black shop-coat where visible
- (2) Camo 9ga HDG hanger nail 1-1/2" 5 lb (~650)
- (3) FastenMaster LedgerLOK 6" (50 ct) — 4x ledger to the house rim, 2 rows staggered 16"
- (66) Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden)
- (1) Black shop-coat for the visible hangers + touch-up paint
- (2) Grace Vycor Deck Protector 12" x 75' (wall membrane behind ledger + side rims)
- (8) Aluminum L-flashing 10' (over ledger + side rims)

HOW

- 4x10 ledger x 55'-2 3/8" total · LedgerLOK 6" 2 rows staggered 16" OC on 71.2 LF of house wall (ledgers + return walls; nothing on a privacy wall) · 107 fasteners
- C: 11 joists @ 12" OC x 22'-5"; B: 13 joists @ 12" OC x 8'-5"; A: 27 joists @ 12" OC x 6'-5". Crown up, hangers at the ledger, both faces of the flush beam.
- 116 x 4x10 blocks, (3) 16d toe-nails each end
- G-Tape 4" on every 4x10 top (joists, rims, ledgers) and the beams; 2" on blocking

CHECK

- Frame square: diagonals within 1/8"
- Every hanger fully nailed; ledger fasteners per the schedule
- Tape continuous; photos of the ledger, hangers and footings before decking



Step 5 · Frame — 4x10 joists at 12", rims, ledger, blocking over the beam

T-405 · Step 6 · Decking

GOES IN

- (18) TimberTech Vintage Coastline 1x6x12 Square Edge
- (76) TimberTech Vintage Coastline 1x6x16 Square Edge
- (2) TimberTech Vintage Coastline 1x6x16 Square Edge — rip boards at the house
- (4) TimberTech Vintage Dark Hickory 1x6x16 Square Edge — borders & dividers
- (2) TimberTech Vintage Dark Hickory 1x6x20 Square Edge — borders & dividers
- (9) Cortex for TimberTech PVC, Coastline plugs (350 ct / 100 SF)
- (1) Cortex for TimberTech PVC, Dark Hickory plugs — borders + dividers

HOW

- Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (2938 field + 152 border/divider)
- Borders first, then the field.
- Rows per zone: C 48 · B 18 · A 14. Borders and dividers first, then the field rows from the outer edge toward the house.

CHECK

- 1/8" gaps held
- No cut end shows at the outer edge
- Color: every board from the same run



Step 6 · Decking — TimberTech Vintage Coastline, Dark Hickory border, hidden fasteners

T-406 · Step 7 · Rail

GOES IN

- (9) TimberTech IRX Cable Rail kit 8' x 36" black (top rail, cable, fittings; cut to bay)
- (10) TimberTech IRX 36" post kit (post + cap + skirt) black
- (4) Drink rail — TimberTech Vintage Dark Hickory 1x6x16 Square Edge laid flat on the top rail, 62.2 LF, mitred at every turn
- (9) TimberTech drink-rail brackets, black — per 8' section (Build SOP: always TimberTech drink-rail brackets)

HOW

- IRX 36" — 10 posts (1 CORNER, 2 END, 7 LINE), 9 bays at 6'-10 3/4" / 6'-2 3/8" / 6'-7" / 7'-0" / 7'-3 5/8" CTC (8' kits cut to bay), posts on the divider lines, no bottom rail; HeadLOK 6" x 4 per post; drink rail board on TimberTech drink-rail brackets, mitred at every turn
- Drink rail on TimberTech drink-rail brackets, mitred at every turn, plugs color-matched.

CHECK

- Top of rail 36" above the decking; posts plumb
- Cables tensioned / balusters under 4" clear
- End of construction



Step 7 · Rail — IRX 36" Matte Black, drink rail on top



The interactive 3D model lives at the online version of this document.

GS Exterior Experts · Deck Division

Prices: rate card · GC at cost · 42.5% GM · financed ÷ 0.93 unless the job says otherwise. Internal — not for the homeowner.